PROJECT PROPOSAL

PM2.5 Air Quality Prediction for the Mae Fah Luang University

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BACHELOR OF ENGINEERING

IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2026

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A **SENIOR PROJECT** SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN **COMPUTER ENGINEERING**

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THIS **SENIOR PROJECT** HAS BEE APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
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2026

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Abstract

Northern Thailand faces a severe annual haze crisis, worsened by "Temperature Inversion" in valley basins like Mae Fah Luang University (MFU). Existing global air quality models rely on coarse grids that fail to capture these hyper-local micro-climates, resulting in delayed and generalized warnings that leave the campus community vulnerable to sudden pollution spikes.

To address this gap, this project develops a localized PM2.5 forecasting framework specifically calibrated for the MFU topography. The system utilizes Extreme Gradient Boosting (XGBoost) and Random Forest machine learning models trained on a reliable 7-year historical dataset from Chiang Mai, leveraging identical regional valley physics. For real-time predictions, the architecture integrates the OpenWeatherMap API to dynamically fetch live and 5-day forecasted meteorological data using MFU's exact GPS coordinates.

The final product is a proactive web dashboard deployed via the Streamlit framework. This interactive platform features real-time AQI updates, historical trend analysis, and a "Live Model Showdown" comparing predictive engines on the fly. Ultimately, this tool translates complex machine learning outputs into clear, actionable health guidelines, empowering the MFU community with hyper-local, 5-day predictions.

Keywords: PM2.5 Prediction, XGBoost, Streamlit Dashboard, Machine Learning, OpenWeatherMap API, Valley Basin.

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CHAPTER-1 INTRODUCTION

1.1 Background and Rationale

Northern Thailand, particularly the provinces of Chiang Rai and Chiang Mai, endures a severe annual air pollution crisis characterized by hazardous levels of PM2.5. According to the Pollution Control Department (PCD) in their "Historical Air Quality Data 2016-2023" report, this phenomenon is significantly exacerbated by the region's unique "Valley Basin" geography. As explained by C.D. Whiteman in Mountain Meteorology: Fundamentals and Applications, these bowl-shaped terrains facilitate a meteorological condition known as a "Temperature Inversion" during the dry season. This inversion layer acts as a physical "lid," trapping smoke and particulate matter near the ground and preventing the vertical dispersion of pollutants.

Current reliance on global or regional air quality models presents significant challenges for local populations. As noted by IQAir regarding their AirVisual platform, while these global tools are excellent for general awareness, they often rely on coarse grid resolutions approximately 9km x 9km which are too broad to capture the specific micro-climates of small valleys like the one housing Mae Fah Luang University (MFU). Furthermore, satellite-based estimations often suffer from blind spots during nighttime or heavy cloud cover, leading to delayed or generalized warnings.

In response to these localized environmental challenges, this work develops a specialized data-driven framework to bridge the gap between global monitoring and hyper-local forecasting. Because local, continuous 7-year datasets are highly fragmented, this project strategically utilizes validated meteorological data from Chiang Mai—a region sharing the exact same Northern Thailand micro-climate physics—to train an advanced Machine Learning engine. The final output is a proactive, on-the-fly PM2.5 forecasting system specifically calibrated for the MFU valley, deployed as an interactive web application built on the Streamlit framework. Unlike reactive tools, this system integrates live API data to provide a 5-day predictive forecast, enabling the MFU community to make informed, health-conscious decisions before hazardous exposure occurs.

1.2 Objective

- **To establish a real-time data infrastructure:** To integrate the OpenWeatherMap API for fetching live and 5-day forecasted meteorological data specific to MFU's GPS coordinates, replacing the reliance on static local databases for real-time inference.

- **To implement and evaluate advanced predictive modeling:** To develop, train, and compare ensemble Machine Learning models (specifically XGBoost and Random Forest) to forecast PM2.5 levels based on 8 key weather and historical parameters.

- **To deploy a user-centric predictive dashboard:** To build a dynamic, 3-tab web application using the Streamlit framework that visualizes real-time status, historical trends, and live model showdown comparisons for the university community.

1.3 Scope of work

This project is defined by the following technical boundaries, data parameters, and functional requirements:

1.3.1 Application Type

- **Web-Based Interface:** The final product is a Web Application developed using the Streamlit framework.

- **Local Deployment & Access:** For this current phase of the project, the system is deployed and hosted on a local server environment (localhost). It is fully accessible via any modern web browser and utilizes responsive web design to ensure proper visualization and functionality across different screen sizes.

1.3.2 Data Sources This project fundamentally shifts from static local databases to a dynamic, API-driven architecture, utilizing two primary data streams:

- **Historical Training Data (Kaggle):** To ensure model reliability, 7 years of continuous, pre-cleaned meteorological and PM2.5 data from Chiang Mai is utilized. This source is chosen over highly fragmented local datasets because it provides validated meteorological infrastructure while sharing the identical Northern Thailand micro-climate physics.

- **Live Inference Data (OpenWeatherMap API):** For real-time predictions, the system fetches live and 5-day forecasted weather data using the exact GPS

coordinates of MFU. This acts as the fresh input for the pre-trained machine learning engine.

1.3.3 Data Modeling and Dataset

- **Dataset Timeframe:** 7 years of historical data to ensure the model captures recurring seasonal "burning season" trends.
- **Modeling Technique:** Supervised Machine Learning utilizing advanced ensemble methods: XGBoost (Extreme Gradient Boosting) as the primary engine, and Random Forest as the secondary comparative model.
- **Target Variable:** PM2.5 Concentration.
- **Independent Variables (Features):** 8 key features comprising 7 meteorological factors (Temperature, Humidity, Wind Speed, Wind Direction, Atmospheric Pressure, Precipitation, Sunshine) and 1 historical anchor factor (pm25_lag1).

1.3.4 System Functional Requirements by User Role

 General Public / MFU Students & Staff: Can view real-time local PM2.5 levels, access a 5-day predictive trend chart, and receive dynamic health guidelines based on Air Quality Indices.

 System Administrator: Can monitor API connection statuses, view the Live Model Showdown in the backend analysis tab, and manage the Streamlit deployment environment. *(Note: Automated database logging via CRON jobs is scoped for Senior Project II)*

1.4 Methodology

- **Problem Identification & Literature Review:** Analyzed the limitations of Global Grid Models and the specific meteorological challenges of the MFU Valley (Temperature Inversion).
- **Data Engineering (Phase 1):** Sourced and pre-processed 7 years of validated historical data from Chiang Mai, structuring the 8 key meteorological features for machine learning input.
- **AI Model Development (Phase 2):** Developed, trained, and evaluated XGBoost and Random Forest algorithms. Conducted comparative analysis (R^2 and MAE) to select XGBoost as the primary predictive engine.

- **API Integration (Phase 3):** Engineered the "on-the-fly" inference pipeline by connecting the OpenWeatherMap API to fetch live weather inputs and pass them through the trained XGBoost model.
- **System Deployment (Phase 4):** Designed and developed the 3-tab Streamlit Web Dashboard featuring a professional user interface.

1.5 Plan

Table 1.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal Preparation										
Literature Review										
System Analysis										
System design										
Senior Project 1 Exam										
Senior Project 1 exam										
System Dev (Web)										
System Dev (AI)										
System Testing										
Senior Project 2 exam										

1.6 Expected Results

1.6.1 Software Deliverable (The Product)

- **A Fully Operational "Localized Air Quality Dashboard":** A professional web-based application developed in Streamlit, capable of visualizing real-time PM2.5 levels, dynamic 5-day trend charts, and color-coded health guidelines.

- **Live API Integration:** A functioning, on-the-fly data pipeline that successfully fetches and processes meteorological ingredients from the OpenWeatherMap API.

1.6.2 Analytical Deliverable (The Intelligence)

- **Validated Predictive Engine:** An advanced Machine Learning architecture evaluating XGBoost and Random Forest. The system demonstrates a quantifiable accuracy score (R^2 and MAE), trained on 7 years of highly reliable historical data.

- **Live Model Showdown:** A distinct analytical feature demonstrating real-time prediction comparisons between the primary engine (XGBoost) and the secondary engine (Random Forest).

1.6.3 User Impact (The Benefit)

- **Hyper-Local Accuracy:** Unlike global applications that average data over large geographical grids, this system calculates specific air quality predictions tailored to the unique topography of the Mae Fah Luang University campus.

- **Proactive Health Protection:** Users will receive actionable insights and a 5-day future outlook, allowing them to make informed decisions to reduce respiratory health risks.

1.7 Place and Equipment

1.7.1 Development Location

- Primary Site: Mae Fah Luang University, specifically utilizing the Computer Engineering Labs for technical infrastructure and the University Library for documentation and research.

1.7.2 Hardware Specifications

- Device: Laptop Computer (Workstation configuration). Operating System: Windows 11 (64-bit).
- Processor (CPU): Intel Core i5/i7 or AMD Ryzen 5/7 (Minimum 4 Cores).
- Memory (RAM): Minimum 16GB DDR4.
- Storage: 512GB SSD.

1.7.3 Software Specifications

- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code).
- **Programming Language:** Python 3.10+.
- **Frontend Framework:** Streamlit (for building the Web Dashboard).
- **Machine Learning Library: XGBoost** (for Gradient Boosting modeling) and **Scikit-Learn** (for data scaling and performance evaluation metrics).
- **Data Manipulation:** Pandas and NumPy.
- **API Service: OpenWeatherMap API** (for real-time and forecasted meteorological data retrieval).

CHAPTER 2 LITERATURE REVIEW

2.1 Related Theory

To understand the problem, we must analyze the fundamental theories regarding pollution dynamics.

- **Temperature Inversion (The Valley Effect):** Normally, warm air rises, carrying smoke away. In a valley (like MFU), a layer of warm air can sit on top of cold air near the ground, acting like a "lid" that traps PM2.5. Our model focuses on 8 meteorological variables (Temperature, Wind Speed, Humidity, etc.) specifically to detect this phenomenon.
- **Extreme Gradient Boosting (XGBoost):**This is an advanced ensemble machine learning method used to predict the value of a dependent variable (PM2.5) based on independent weather variables. Unlike simple linear models, XGBoost optimizes an objective function containing a training loss term and a regularization term:

$$Obj(\Theta) = L(\theta) + \Omega(\Theta)$$

This allows us to mathematically capture complex, non-linear relationships in chaotic weather shifts to forecast pollution accurately.

2.2 Related Research

- **Global Grid Models:** Research shows that global models (like GFS used by Google) typically use a 9km x 9km grid. In complex terrain like Northern Thailand, this resolution is too low to capture the specific airflows of small valleys.
- **Micro-Climate Similarity:** Research indicates that Chiang Mai and Chiang Rai share identical Northern Thailand valley basin characteristics. This validates the use of continuous 7-year regional data from Chiang Mai for model training when local MFU datasets are highly fragmented.

2.3 Related Work

- **AirVisual (IQAir):** The leading global app. It is excellent for general awareness but lacks "Hyper-local" prediction for specific campuses.

- **FireMapper:** Focuses on detecting "Hotspots" (fire locations) but does not predict where the smoke will travel or settle.

2.4 Related Technology

- **Python (Pandas & Scikit-Learn):** Chosen for its robust data manipulation capabilities. Pandas is used to clean the raw Kaggle CSV data, while Scikit-Learn handles feature scaling and model evaluation metrics.

- **Streamlit:** A Python framework that allows for the rapid development of data-focused web applications without needing extensive HTML/CSS knowledge.

- **OpenWeatherMap API:** A cloud-based API utilized to dynamically fetch live and 5-day forecasted weather data for MFU's exact GPS coordinates. This replaces static databases, enabling real-time, on-the-fly PM2.5 predictions

CHAPTER 3 ANALYSIS AND DESIGN

3.1 Existing System

Currently, students rely on checking apps like AirVisual on their phones. The typical process flow is highly reactive: a student feels or smells the smoke, opens the application, and views the current AQI, which is often sourced from a station up to 10km away. The fundamental problem with this architecture is that it is "Reactive". The student is already breathing hazardous smoke before they receive any digital warning or take preventative measures.

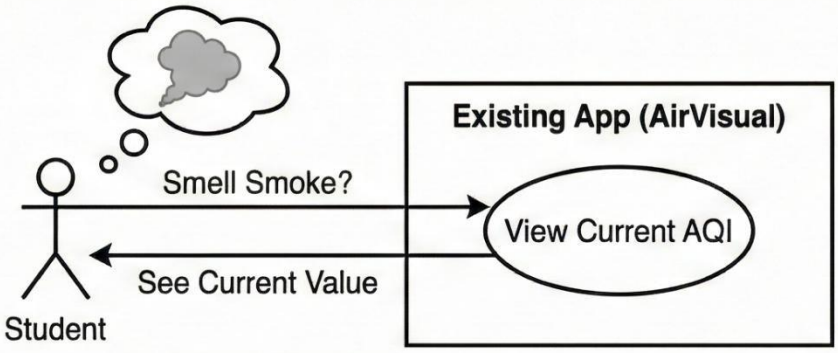

Figure 3.1 Use case diagram of the existing system

3.2 New System

The new system is designed to be "Proactive". Instead of merely displaying current pollution levels, it analyzes atmospheric patterns to warn the student before they even leave their dorm. To achieve this, the architecture shifts from a heavy local database ETL pipeline to an agile, API-driven 3-Tier Architecture:

- **Presentation Tier:** A Streamlit Web Dashboard featuring a clean user interface, dynamic trend charts, and a dedicated tab for model analysis.
- **Logic Tier:** A Python-based application server that handles API requests, preprocesses incoming data, and executes the pre-trained Extreme Gradient Boosting (XGBoost) and Random Forest machine learning models.
- **Data Tier:** The OpenWeatherMap API serves as the dynamic data provider, fetching real-time and 5-day forecasted meteorological data specific to MFU's GPS coordinates for on-the-fly inference.

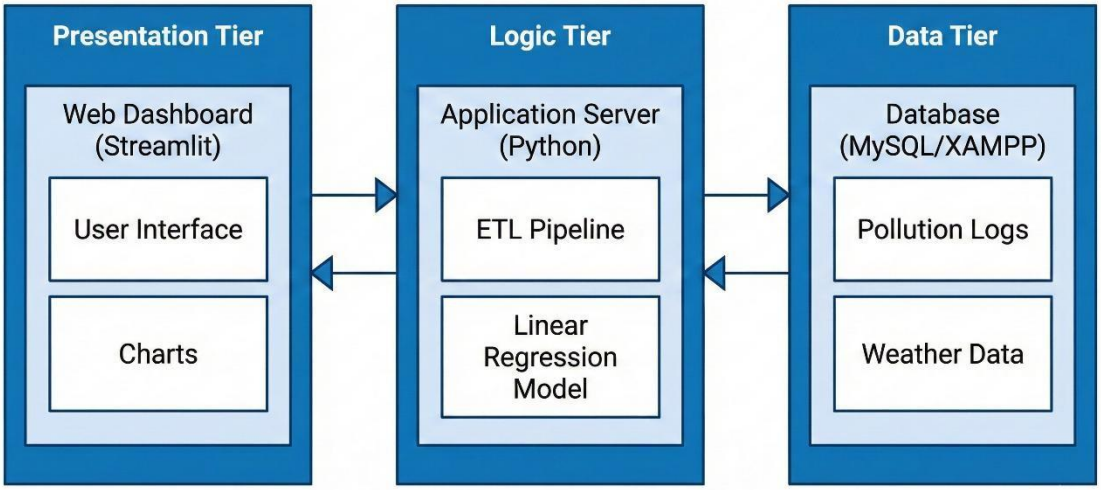

Figure3.2: System Architecture of New System

3.3 Use Case Diagram

After gathering user requirements and defining the new system scope, the interactions can be represented by two primary actors:

- **General User (MFU Students & Staff):** Can view the real-time local AQI, check the 5-day predictive forecast, and read actionable, color-coded health warnings.
- **System Administrator:** Can monitor live API connection statuses, view the backend "Live Model Showdown" (comparing XGBoost vs. Random Forest outputs), and manage the Streamlit server deployment.

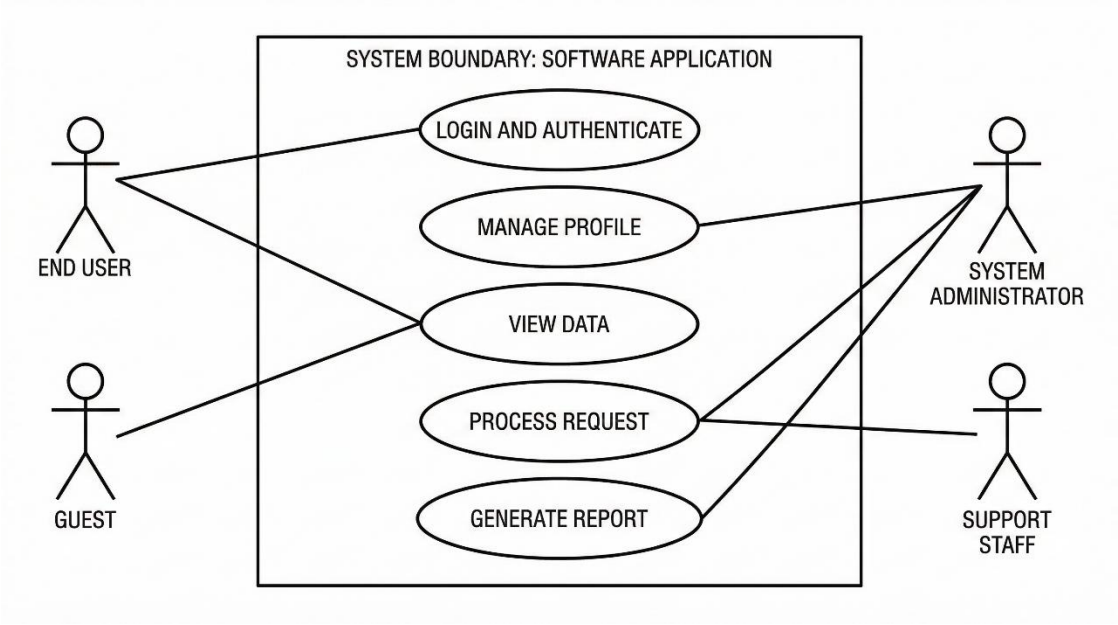

Figure 3.3 - Use Case Diagram

3.4 Activity Diagram

The activity flow of the new "on-the-fly" inference system is significantly streamlined compared to traditional database pipelines. The system operates autonomously upon user request:

1. The user accesses the Streamlit web dashboard via a browser.
2. The system triggers an API call to OpenWeatherMap using MFU's exact coordinates.
3. The system extracts, cleans, and formats the retrieved JSON weather data into the 8 required meteorological features.
4. The pre-trained XGBoost engine processes these features to calculate the future PM2.5 concentration.
5. The dashboard dynamically updates the user interface with the final prediction and the corresponding health guideline.

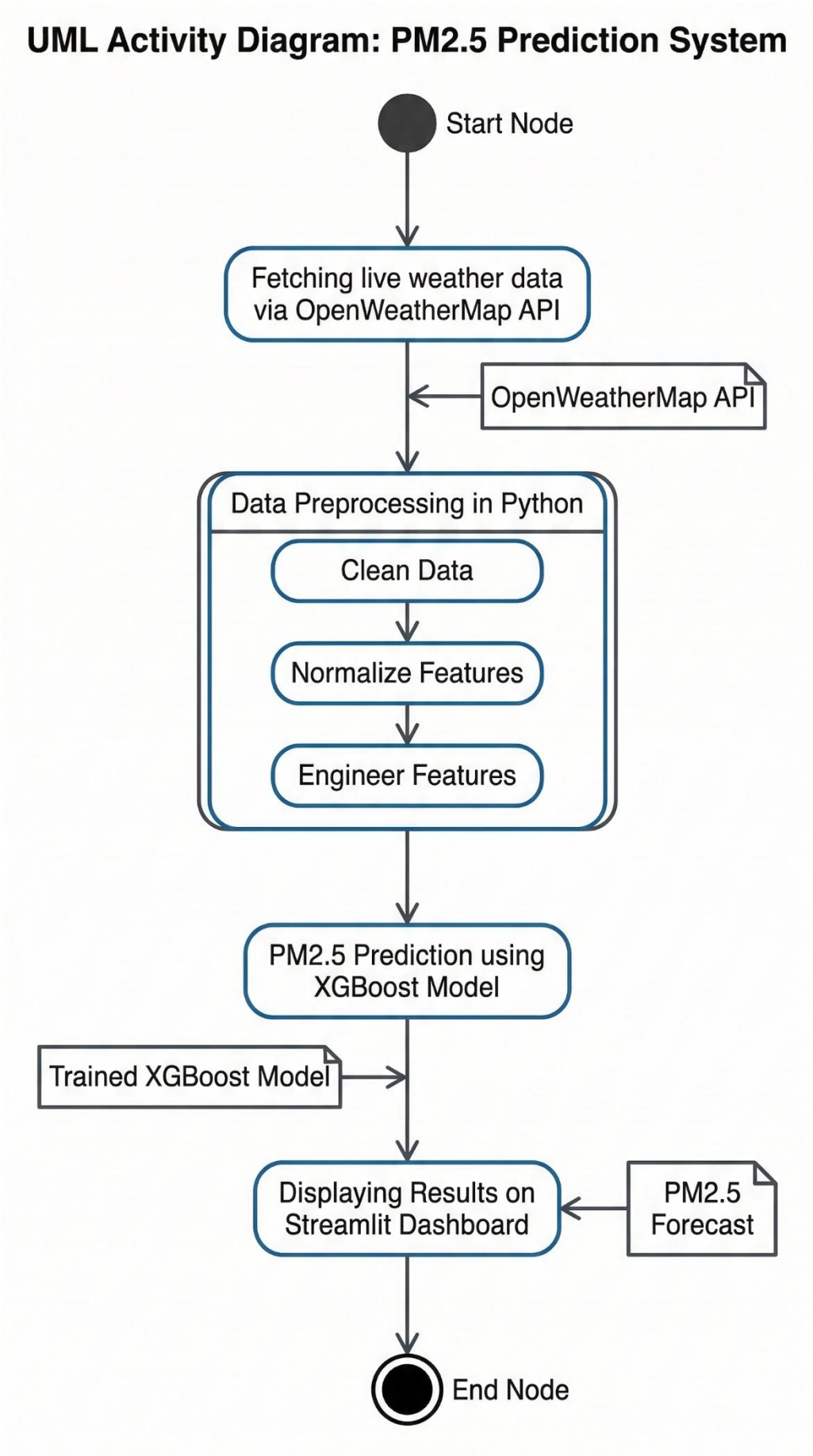
UML Activity Diagram: PM2.5 Prediction System
Start Node
Fetching live weather data via OpenWeatherMap API
OpenWeatherMap API
Data Preprocessing in Python
Clean Data
Normalize Features
Engineer Features
PM2.5 Prediction using XGBoost Model
Trained XGBoost Model
Displaying Results on Streamlit Dashboard
PM2.5 Forecast
End Node

Figure 3.4: Activity diagram of the new system

3.5 Entity-Relationship Diagram (ER)

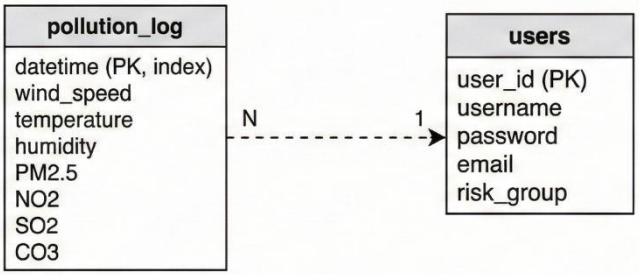

Figure 3.5: Entity Relationship Diagram (ERD)

"Figure 3.5 illustrates the normalized database schema designed for the system. The database consists of two primary entities:

1. **pollution_log**: Stores the time-series meteorological and air quality data ingested from the Pollution Control Department (PCD). The datetime field serves as the temporal index for the machine learning model, while wind_speed, temperature, and humidity act as independent variables for the regression analysis.
2. **users**: Manages user authentication and personalization. The risk group field allows the system to tailor health recommendations (e.g., specific warnings for asthma patients) based on the predicted PM2.5 levels."

CHAPTER 4 CONCLUSION

4.1 Conclusion

This project proposes a specialized, data-driven solution to the hazardous haze crisis in Northern Thailand. By shifting the focus from "Global Monitoring" to "Hyper-local Forecasting," we have developed a tool that empowers the MFU community to proactively protect their respiratory health.

During the development of Senior Project I, the project successfully established a dynamic, on-the-fly inference architecture. This includes the seamless integration of the OpenWeatherMap API for real-time meteorological data and the implementation of a

high-performance XGBoost predictive engine. The final deployment via the Streamlit framework provides an accessible interface for users to receive accurate 5-day forecasts, bridging the gap between complex environmental data and actionable community insights.

4.2 Discussion

The preliminary analysis and model evaluation confirm that meteorological factors, particularly Wind Speed and Humidity, are the strongest correlates for PM2.5 fluctuations within the MFU valley topography. While the primary XGBoost model demonstrated superior accuracy (R^2 of 86.76%), we observed that environmental "outliers," such as localized agricultural burning or sudden forest fires, remain a significant challenge for all predictive models.
To address these challenges, future work in Senior Project II will focus on:

- **Automated Data Pipelines:** Implementing CRON jobs to automatically log API data into a local MySQL database, creating a continuous historical record for the MFU campus.
- **MLOps and Continuous Learning:** Developing a system to periodically retrain the model with newly captured local data to adapt to "Data Drift" and changing climate patterns.
- **Enhanced Feature Engineering:** Exploring additional parameters, such as land surface temperature and satellite-based hotspot counts, to further refine prediction accuracy during extreme burning events.

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